Proof. It suffices to prove this for two summands. If α_i are eigenvalues of rational matrices A_i of smallest size (i.e., their characteristic polynomials are the minimal polynomials of α_i), then $\alpha_1 + \alpha_2$ is an eigenvalue of $A := A_1 \otimes \mathrm{Id} + \mathrm{Id} \otimes A_2$. Therefore, so is any algebraic conjugate to $\alpha_1 + \alpha_2$. But all eigenvalues of A are of the form $\alpha_1' + \alpha_2'$, so we are done. $\square$

Problem 5.2.7. (a) Show that for any finite group G there exists a finite Galois extension $K \subset \mathbb{C}$ of $\mathbb{Q}$ such that any finite dimensional complex representation of G has a basis in which the matrices of the group elements have entries in K.

Hint: Consider the representations of G over the field $\overline{\mathbb{Q}}$ of algebraic numbers.

(b) Show that if V is an irreducible complex representation of a finite group G of dimension > 1, then there exists $g \in G$ such that $\chi_V(g) = 0$.

Hint: Assume the contrary. Use orthonormality of characters to show that the arithmetic mean of the numbers $|\chi_V(g)|^2$ for $g \neq 1$ is < 1. Deduce that their product β satisfies $0 < \beta < 1$. Show that all conjugates of β satisfy the same inequalities (consider the Galois conjugates of the representation V, i.e., representations obtained from V by the action of the Galois group of K over $\mathbb{Q}$ on the matrices of group elements in the basis from part (a)). Then derive a contradiction.

Remark 5.2.8. Here is a modification of this argument, which does not use (a). Let $N = |G|$. For any $0 < j < N$ coprime to N, show that the map $g \mapsto g^j$ is a bijection $G \to G$. Deduce that $\prod_{g \neq 1} |\chi_V(g^j)|^2 = \beta$. Then show that $\beta \in K := \mathbb{Q}(\zeta)$, $\zeta = e^{2\pi i/N}$, and that it does not change under the automorphism of K given by $\zeta \mapsto \zeta^j$. Deduce that β is an integer, and derive a contradiction.

5.3. Frobenius divisibility

Theorem 5.3.1. *Let G be a finite group, and let V be an irreducible representation of G over $\mathbb{C}$. Then*

$$\dim V \text{ divides } |G|.$$